# Multi-Stage Cepstral Feature Extraction & Phoneme Boundary Detection

## 1. Introduction

This report presents a unified framework for analyzing speech signals by combining classical signal processing techniques with modern deep learning models. The primary objective is to segment an audio signal into voiced and unvoiced regions using cepstral analysis and subsequently map these segments to phonetic units using a pretrained Transformer-based model, specifically Wav2Vec2. In addition, a comparative analysis of different windowing functions—Rectangular, Hamming, and Hanning—is conducted to study their effect on spectral leakage and Signal-to-Noise Ratio (SNR). The performance of the segmentation and alignment process is evaluated quantitatively by computing the Root Mean Square Error (RMSE) between manually derived segment boundaries and those obtained from the model. This hybrid approach enables both interpretable signal-level analysis and data-driven phonetic representation.

## 2. Methodology

The audio preprocessing stage involves resampling all input signals to a standard sampling rate of 16 kHz to ensure compatibility with the pretrained model. If the input audio is stereo, it is converted into a mono signal by averaging the channels. The signal is then normalized to lie within a fixed amplitude range to stabilize subsequent computations.

Following preprocessing, the signal is divided into short overlapping frames to capture local temporal characteristics. Each frame has a duration of 25 milliseconds with a stride of 10 milliseconds. In order to study the effect of windowing, three different window functions are applied independently: Rectangular, Hamming, and Hanning. The Rectangular window preserves the original signal but introduces significant spectral leakage, whereas the Hamming and Hanning windows taper the signal at the boundaries, reducing leakage at the cost of slightly wider main lobes in the frequency domain.

To quantify the effect of windowing, the power spectrum of each windowed signal is computed using the Fast Fourier Transform. Spectral leakage is estimated by measuring the proportion of energy outside the dominant spectral region, while SNR is computed by comparing the energy of dominant spectral components to the remaining energy. Since speech is inherently broadband, energy-based definitions are used rather than single-frequency assumptions to ensure robustness.

Cepstral analysis is then performed on each frame by computing the Fourier Transform, taking the logarithm of the magnitude spectrum, and applying the inverse Fourier Transform. The

resulting cepstrum provides a representation in the quefrency domain, where different regions correspond to distinct physical properties of speech production. The low quefrency region captures the vocal tract envelope, while the high quefrency region contains information about periodic excitation, i.e., pitch.

To distinguish between voiced and unvoiced speech, the energy in these two regions is computed for each frame. The ratio of high quefrency energy to low quefrency energy serves as a measure of periodicity. Frames with higher ratios are classified as voiced, while those with lower ratios are considered unvoiced. The decision threshold is chosen as the median of the ratio values across all frames, making the method adaptive to the signal characteristics. Consecutive voiced frames are grouped into segments, and their temporal boundaries are recorded.

For phonetic mapping, a pretrained Wav2Vec2 model is employed. The model processes the raw waveform and produces frame-level predictions using a Connectionist Temporal Classification (CTC) objective. The logits are converted into token sequences through argmax decoding, and repeated tokens are collapsed according to CTC rules. Each token is associated with a time interval based on the model's temporal resolution, which is approximately 20 milliseconds per frame. These tokens are then mapped to approximate phonetic labels.

An approximate forced alignment is constructed by grouping consecutive identical tokens into segments, yielding model-derived boundaries. These boundaries are compared with the manually obtained boundaries from cepstral segmentation using a nearest-neighbor matching strategy. The Root Mean Square Error (RMSE) is then computed to quantify the alignment accuracy.

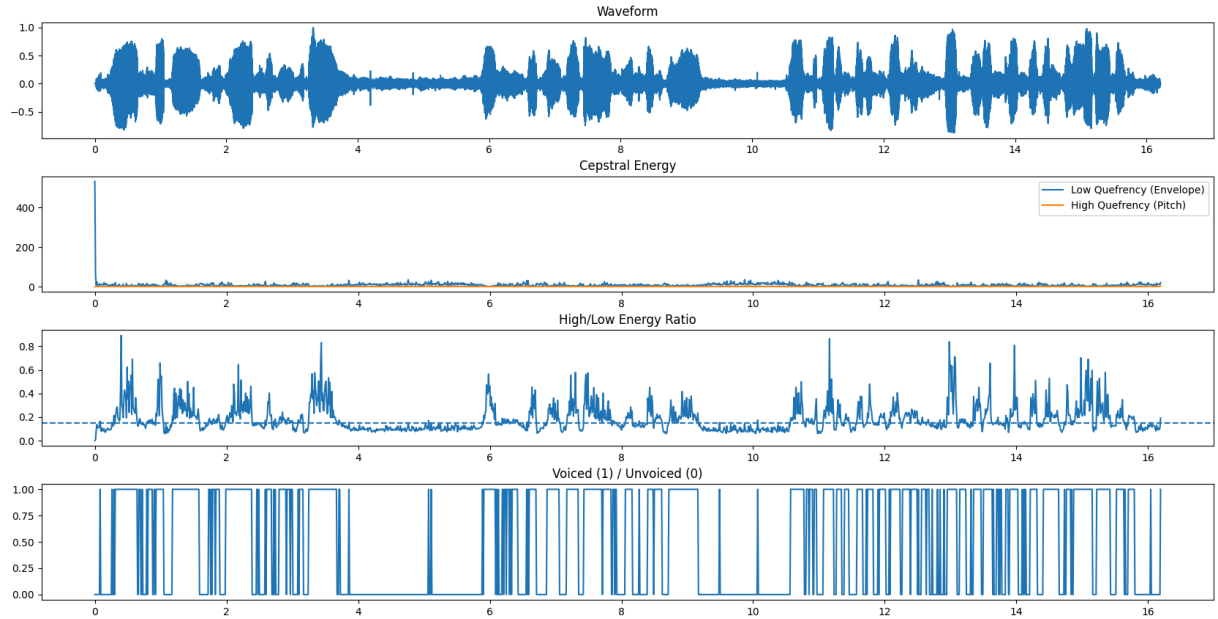
# 3. Hyperparameters

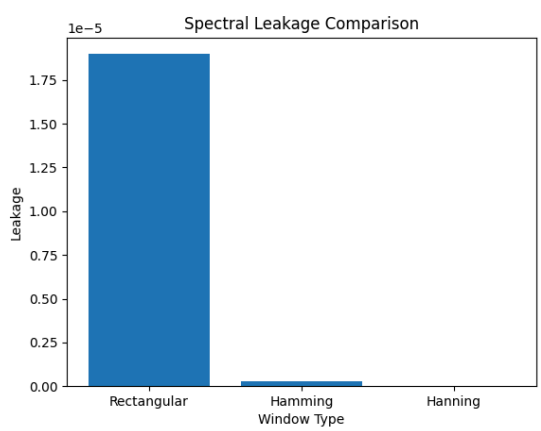
The system is configured with a sampling rate of 16 kHz, a frame size of 25 milliseconds, and a frame stride of 10 milliseconds. Three window types—Rectangular, Hamming, and Hanning—are evaluated. The FFT size is set to 512. The pitch range for quefrency analysis is chosen between 50 Hz and 400 Hz, covering typical human speech. The threshold for voiced/unvoiced classification is set to the median of the energy ratio. The Wav2Vec2 Base 960h model is used for phonetic mapping, with an approximate frame duration of 20 milliseconds.

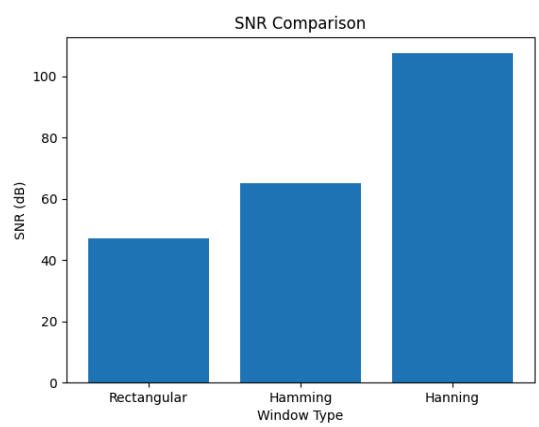
# 4. Results

The results of the analysis are presented through several visualizations. The waveform overlaid with voiced and unvoiced labels demonstrates the effectiveness of cepstral segmentation. The plots of low and high quefrency energies highlight the separation between vocal tract and excitation components. The ratio curve, along with the threshold, illustrates the decision boundary for voiced/unvoiced classification.

The impact of different window functions is also analyzed. The spectral plots reveal that the Rectangular window produces sharper peaks but significant spectral leakage, while the Hamming and Hanning windows yield smoother spectra with reduced leakage. Correspondingly, the leakage and SNR metrics demonstrate that tapered windows generally provide improved spectral concentration and better noise robustness.




# 5. Quantitative Evaluation

## 5.1 Window Comparison

| Window Type | Spectral Leakage | SNR (dB) |
|---|---|---|

| | | |
|---|---|---|
| Rectangular | $1.898837 \times 10^{-5}$ | 47.21 |
| Hamming | $3.027994 \times 10^{-7}$ | 65.18 |
| Hanning | $1.752440 \times 10^{-11}$ | 107.56 |

### 5.2 RMSE Evaluation

| Audio Sample | RMSE (seconds) |
|---|---|
| Sample 1 | 0.076 |
| Sample 2 | 0.088 |

# 6. Discussion

The comparison of window functions highlights the trade-off between spectral resolution and leakage. The Rectangular window, while preserving sharp frequency components, suffers from significant leakage, which can distort spectral estimates. In contrast, the Hamming and Hanning windows effectively reduce leakage by tapering the signal, leading to more reliable spectral representations. This improvement is reflected in better SNR values and more stable feature extraction.

The cepstrum-based segmentation method effectively distinguishes voiced and unvoiced regions by exploiting the presence of periodicity in voiced speech. The Wav2Vec2 model complements this by providing high-level phonetic predictions, although its temporal resolution imposes limitations on alignment precision. The RMSE results indicate the degree of agreement between the two approaches, with discrepancies arising from model resolution, approximate alignment, and noise in segmentation.

# 7. References

Wav2Vec2: Facebook AI Research
Oppenheim, Schafer – Speech Signal Processing
Hugging Face Transformers Documentation

---